

Crypto Perpetual Futures vs Stocks for Systematic Strategy Testing

Executive summary

A stock backtest can often survive with one price series, simple commissions, point-in-time splits/dividends, and a single delisting rule. A crypto perpetual futures backtest cannot. Across Binance, Bybit, OKX, Coinbase International Exchange, Deribit, and Hyperliquid, the exchange risk engine uses a venue-defined mark price rather than last trade for unrealized PnL and liquidation; funding is a recurring cashflow on notional, not on margin; liquidation is venue-specific and may be staged, partial, or delegated to insurance-fund / ADL / backstop mechanisms; and instrument lifecycle data is not optional because pre-launch states, halts, delistings, automatic settlement, and symbol metadata all affect strategy viability and realized returns. Treating a perp like a stock or even like a standardized CME future will usually misstate both edge and blow-up risk. ¹

For small-account strategies, the most dangerous assumptions are usually: ignoring funding, simulating liquidation off last price, collapsing maker/taker into a flat fee, assuming stops always fire before liquidation, ignoring staged liquidations, and assuming contracts have no lifecycle events. Even “small” recurring costs scale fast because they apply to notional. For example, Binance’s default interest component is 0.01% per 8-hour funding interval; on a \$10,000 notional position, that is \$1 per interval or about \$3 per day before the premium component. On a 10x leveraged account backed by \$1,000 margin, that baseline alone is about 0.3% of account equity per day and roughly 9% of margin over 30 days if sustained. A two-taker round trip at 5 bps each side costs about 10 bps before slippage; that is \$10 per \$10,000 round trip, and repeated turnover quickly dominates small expected alpha. ²

The highest-confidence implementation rule is simple: every exchange needs at least four distinct time series in the backtest state machine—last/trade, mark, index/oracle, and funding—plus fee-tier history, order-type semantics, liquidation rules, and contract lifecycle metadata. If you only have OHLCV and a flat bps fee, you do not yet have a defensible crypto perp backtest. ³

What makes crypto perpetuals structurally different from stocks

Stocks usually have no native mark-price liquidation engine, no periodic funding exchange between longs and shorts, and no maker/taker microstructure embedded into the contract economics. Crypto perps do. The exchange itself defines which price triggers liquidation, how funding is computed and settled, what happens when liquidation cannot be completed cleanly, and how stop orders interpret the trigger price. Those rules vary enough across venues that a “generic perp simulator” is often wrong in exchange-specific ways. ⁴

The most important practical difference from stocks is that PnL accounting and solvency accounting diverge. The last trade may be the best input for entry/exit slippage modeling, but mark price is typically the

input for unrealized PnL, margin checks, and liquidation. Binance states this directly for futures; Bybit does the same; OKX defines mark price as the reference for unrealized PnL and forced liquidation; Coinbase International Exchange uses a central-limit-order-book mark for unrealized PnL and liquidation checks; Deribit's mark price is a reference price for unrealized PnL, margin, and liquidation; Hyperliquid's mark price is used for margining, liquidations, TP/SL triggering, and unrealized PnL. ¹

Funding is the second large difference. It is neither a commission nor an exchange fee; it is a transfer between longs and shorts computed off exchange-defined pricing inputs and assessed at exchange-defined times. The platform often acts only as counterparty router or settlement facilitator. On Binance, funding payments go directly between opposing traders. OKX says the same. Bybit updates the displayed funding rate every minute. Coinbase International Exchange settles hourly. Deribit accrues funding continuously in minute fractions of the 8-hour rate. Hyperliquid computes an 8-hour-style rate but pays one-eighth every hour. These differences materially affect holding-period returns and therefore strategy ranking. ⁵

Comparative exchange matrix

Funding rates and mark price construction

Exchange	Funding calculation and settlement timing	Mark price construction and its role	Sources
Binance	Funding amount = nominal position value × funding rate, with nominal value based on mark price . Default interest component is 0.03% daily = 0.01% per 8h . Funding periods are typically 8h and Binance notes a 15-second deviation can occur around the timestamp. The premium component is computed from impact prices and a weighted average premium index. ⁶	USD [Ⓢ] -M perpetuals: Mark = Median(Price 1, Price 2, Contract Price) , with Price 1 = Index × [1 + Last Funding Rate × time-until-next-funding / funding-period] and Price 2 = Index + 30-second basis moving average , where the basis uses 1-second samples. Unrealized PnL and liquidation are mark-price based; in extreme conditions Binance may fall back to Price 2 and can place a symbol into reduce-only mode. ⁷	⁸

Exchange	Funding calculation and settlement timing	Mark price construction and its role	Sources
Bybit	Funding rate has interest rate + average premium index ; Bybit recalculates the components every minute and the published rate updates every minute. The premium index uses impact bid/ask vs index; the average premium uses a weighted average over minute samples in the window. Funding intervals are symbol-specific via API metadata; common instruments show fundingInterval = 480 minutes. Pre-market perps use different rules. ⁹	Standard perpetuals: Mark = Median(Price 1, Price 2, Last Traded Price) , with Price 1 = Index × [1 + Last Funding Rate × (Time Until Funding / 8)] and Price 2 = Index + 2.5-minute basis moving average sampled every second. If spot-index data is abnormal or insufficient, Bybit can fall back to last traded price . Mark price is used for unrealized PnL and liquidation, not actual realized execution PnL. ¹⁰	¹¹
OKX	Default settlement is every 8 hours at 00:00, 08:00, 16:00 UTC , but OKX can change intervals to 1/2/4/6/8h depending on market conditions. Current formula: Funding = clamp[(Average premium index + clamp(Interest – Average premium index, ±0.05%)) / (8/N), cap, floor] , where interest = 0.01% fixed and the premium index uses impact bid/ask vs index. The rate used at assessment is the one computed in the previous minute . ¹²	Mark = Index price + moving-average basis . OKX uses mark price for unrealized PnL and to reduce unnecessary forced liquidation from abnormal last trades. Public API exposes mark price separately; WebSocket mark-price candles are also available. ¹³	¹⁴
Coinbase International Exchange	Funding is hourly . Official INTX help defines Funding Rate = TWAP(Premium, 1 hour, 1 second) × α + Previous Funding Rate × (1–α) , where Premium = (Mark – Index) / Index / 24, smooth = 6 , and the default lookback parameter shown is n = 7 . Funding fee = Position Value × Funding Rate . ¹⁵	FIX market-data docs state mark price is the median of best bid, best offer, and last trade . INTX help says mark-to-market uses the central limit order book mark price for unrealized PnL, margin checks, and liquidation triggers. ¹⁶	¹⁷

Exchange	Funding calculation and settlement timing	Mark price construction and its role	Sources
Deribit	Official support explains perp funding through a premium rate based on (mark – index)/index , with a capped formula around a $\pm 0.025\%$ band, and funding is effectively assessed in minute fractions of the 8-hour rate . The worked examples explicitly divide by 480 minutes for a 1-minute holding period. API market data exposes both current_funding and funding_8h . ¹⁸	Official support states mark price is the reference for unrealized PnL, margin, and liquidation. Support material describes it as index price plus a 30-second EMA-type fair-value adjustment , with a last-trade / best-bid-ask contingency in thin conditions. API ticker and book endpoints publish mark price. ¹⁹	²⁰
Hyperliquid	Funding is paid every hour . Official formula is an 8-hour-style formula paid in 1/8 increments each hour: F = Average Premium Index + clamp(interest – P, ± 0.0005) ; premium is sampled every 5 seconds and averaged over the hour; funding is capped at 4% per hour . Hyperliquid notes this is closer to centralized-exchange funding than typical onchain designs. ²¹	Hyperliquid calls mark price an “unbiased and robust estimate” used for margining, liquidations, TP/SL triggers, and unrealized PnL . Mark/oracle are updated roughly every 3 seconds when validators publish oracle prices. Liquidations explicitly use mark, not raw book price. ²²	²³

Liquidation, insurance, ADL, and fee treatment

Exchange	Liquidation trigger and mechanics	Trading fees, rebates, and liquidation-fee treatment	Sources
Binance	Liquidation is triggered when mark price reaches liquidation price. Binance warns the actual liquidation mark can be materially different from the displayed threshold in fast markets; its example shows liquidation displayed at 17,006 but executed at mark 17,100 after a jump. ADL is the final step if the insurance fund cannot absorb the bankrupt position. ²⁴	Regular examples show 0.02% maker / 0.05% taker for both USD [Ⓢ] -M and COIN-M illustrations. Exchange metadata includes a symbol-specific liquidationFee field. Stop-like orders can be triggered on <code>MARK_PRICE</code> or <code>CONTRACT_PRICE</code> and can optionally require the mark/contract spread to be within <code>triggerProtect</code> . ²⁵	²⁶

Exchange	Liquidation trigger and mechanics	Trading fees, rebates, and liquidation-fee treatment	Sources
Bybit	Liquidation uses mark price. Isolated liquidation-price formulas explicitly include estimated taker fee to close and risk-tier maintenance margin. UTA liquidation is staged: if earlier steps restore MMR below 100%, liquidation halts; at high-risk states the engine can liquidate the entire position. Insurance fund covers deficits; ADL is used if insurance is insufficient. 27	Official fee examples for perp contracts show 0.055% taker / 0.02% maker. Delisted positions are closed with a 0.05% fixed settlement fee rather than normal VIP trading fees. I did not retrieve a separate public document stating an additional liquidation fee beyond the close-fee treatment embedded in liquidation formulas. 28	29
OKX	Forced liquidation begins when mark price reaches forced-liquidation price and margin falls below requirement. OKX distinguishes forced-liquidation price, bankruptcy price, and the entrusted price the liquidation engine uses. Public/private API surfaces also expose liquidation-order and ADL-warning channels. OKX help material outside the main liquidation note also describes partial liquidation behavior in some scenarios. 30	Official futures examples use 0.02% maker / 0.05% taker. Forced liquidation fees are charged at the user's current taker fee level. Expiry settlement fee is 0.01%. 31	32

Exchange	Liquidation trigger and mechanics	Trading fees, rebates, and liquidation-fee treatment	Sources
Coinbase International Exchange	INTX defines Initial Margin, Maintenance Margin, and Close-out Margin. If current margin falls below MM, the portfolio is partially liquidated back toward IM. If it falls below CoM, the risk waterfall escalates: check other eligible funds, assign to Liquidity Support Providers, then use the Insurance Fund. Help docs describe automatic liquidation management and order cancellation. 33	INTX has official International Exchange Fees and Liquidity Program pages, but I did not extract the exact tier table into this report. What is clear from the retrieved docs is that fees and maker-liquidity incentives are exchange-specified and should be modeled from the venue's fee page, not assumed from equities conventions. 34	35
Deribit	Deribit liquidations differ by margining regime, but the general rule is maintenance-margin failure and liquidation by the risk engine. Support states liquidation trades carry an additional liquidation fee assigned to the insurance fund. If insurance cannot absorb losses, ADL-like mechanisms can apply depending on regime and product. 36	Support fee page states BTC/ETH inverse futures and perps: 0.00% maker / 0.05% taker; USDC perps and futures: 0.00% maker / 0.05% taker in the examples shown. API instrument metadata also exposes max_liquidation_commission for instruments. 37	38

Exchange	Liquidation trigger and mechanics	Trading fees, rebates, and liquidation-fee treatment	Sources
Hyperliquid	Liquidation uses mark price . Most liquidations are sent directly to the order book; during backstop liquidation , maintenance margin is not returned. Hyperliquid explicitly says this design is different from many CEX designs and that there is no liquidation clearance fee . If account or isolated-position value becomes negative, ADL ranks opposite-side traders by a published formula. ³⁹	Hyperliquid has an official Fees page with fee tiers and special adjustments for aligned quote assets, and builder-code fees can be added on a per-order basis. In the retrieved material, I did not extract the full base tier schedule numerically, so those exact rates should be pulled directly from the fee page before production modeling. What is explicit is that some quote-asset regimes get 20% lower taker fees and 50% better maker rebates . ⁴⁰	⁴¹

Open interest, contract lifecycle, and stop behavior

| Exchange | Open interest reporting and timestamp granularity | Listing, contract status, delisting, historical mapping | Exchange-specific stop behavior | Sources | --- | --- | --- | --- | | **Binance** | Public REST returns **present open interest** for a symbol with a **transaction time** field. ⁴² | `exchangeInfo` includes **onboardDate**, **deliveryDate**, **status**, and `contractType`; for backtests this is the minimum mapping layer for symbol birth and exchange-known status. ⁴³ | Conditional orders can trigger off `MARK_PRICE` or `CONTRACT_PRICE` via `workingType`. With `priceProtect=true`, triggering is blocked if mark-vs-contract spread exceeds the symbol's `triggerProtect`. This matters because liquidation is still mark-price based. ⁴⁴ | ⁴⁵ | | **Bybit** | OI API supports historical buckets of **5m**, **15m**, **30m**, **1h**, **4h**, **1d** with millisecond timestamps; query history goes back to the **symbol launch time**. ⁴⁶ | Instrument metadata includes **launchTime**, **deliveryTime** (for perps, used as **delisting time**), **status**, and `fundingInterval`. Pre-market instruments appear as `PreLaunch`. ⁴⁷ | Bybit conditional orders can use **Last Traded Price**, **Mark Price**, or **Index Price** as trigger basis. Help docs explicitly warn that a stop using LTP may not fire before a **mark-price liquidation**. ⁴⁸ | ⁴⁹ | | **OKX** | Public API returns OI in **contracts**, **coin**, and **USD**, with millisecond timestamp; WebSocket pushes OI **every 3 seconds when updated**. ⁵⁰ | Instruments metadata includes `listTime`, `expTime`, and `state`. Docs say symbols can be `preopen`, then `live`; when delisting is announced, `expTime` is updated. Delisting help says open positions are automatically settled, no fee is charged on delisting, and funding at 16:00 on delisting day is set to **0**. ⁵¹ | OKX stop/TP-SL triggers can use **last price**, **mark price**, or **index price**. Entire-position TP/SL places a market order when triggered; partial TP/SL can be market or limit. ⁵² | ⁵³ | | **Coinbase International Exchange** | Retrieved official public docs expose OI in market data as FIX **MDEntryType = C**, described as the total number of active positions, with **TransactTime** timestamps. I did not retrieve a separate public REST OI-history endpoint. ⁵⁴ | FIX instrument-definition messages provide **SecurityUpdateAction = Add/Modify/Delete**, `LastUpdateTime`, market status states (including `Pre-open`, `Pause`, `Ready to trade`, `Not traded on this market`), and pre-launch conversion events. This is enough to build a lifecycle state machine if you store security definitions over time. Coinbase also states INTX launched on **2 May**

2023. ⁵⁵ | I did **not** retrieve an INTX doc showing mark- vs index-trigger selection like Bybit/OKX/Binance. The official TPSL doc says regular stop orders are **not live until the price is traded through StopPx**, and TPSL orders have different live/reprice behavior. That implies trade-based trigger semantics in at least that workflow. ⁵⁶ | ⁵⁷ | | **Deribit** | API book/ticker endpoints publish `open_interest`; docs specify units differ by product: **inverse futures/perps in USD amount units, linear futures/options in base coin units**. ⁵⁸ | `public/get_instruments` and `public/get_instrument` expose **creation_timestamp**, **expiration_timestamp**, `state`, `is_active`, commissions, and product metadata. The `instrument.creation.*` subscription fires **once at creation time** and carries the same metadata, allowing precise point-in-time symbol birth tracking. ⁵⁹ | Official stop-limit docs allow **mark price, index price, or last price** as trigger reference. ⁶⁰ | ⁶¹ | | **Hyperliquid** | Official API `assetCtxs` includes **mark price, current funding, open interest, etc**. Historical data archives also include **asset contexts**, but Hyperliquid notes S3 uploads are only approximately monthly and may be missing. I did not retrieve an official definition of the OI unit beyond the API surface. ⁶² | For builder-deployed perps, `haltTrading` can **cancel all orders and settle positions to the current mark price**, and can later resume trading, effectively recycling the asset. For the main perp venue, I did not retrieve the full official delisting page contents in this run. ⁶³ | Hyperliquid TP/SL uses **mark price** as trigger. TP/SL market orders have **10% slippage tolerance**. Trigger orders also face user open-order limits and reduce-only/trigger-order rejection conditions at high order counts. ⁶⁴ | ⁶⁵ |

Backtest requirements checklist

A defensible perp backtest needs a richer event model than a stock backtest. The minimum durable schema is the following. Each item below exists because at least one of the venues above uses it directly in funding, liquidation, order triggering, or lifecycle management. ⁶⁶

Core market data

- Last trade / trade tape with exchange timestamps.
- Best bid and best ask, or full depth if you want exchange-faithful impact prices and stop/trigger logic.
- Mark price series.
- Index price or oracle price series.
- Funding-rate snapshots and funding settlement timestamps.
- Open interest snapshots with the venue's own units and normalization metadata.
- Instrument metadata snapshots at every change event, not just the latest schema. ⁶⁷

Funding application logic

- Apply funding on the exchange's schedule, not on candle boundaries. Binance highlights timestamp slippage of around 15 seconds; OKX uses the most recently calculated previous-minute rate; Coinbase INTX is hourly; Deribit examples accrue in minute fractions of the 8-hour rate; Hyperliquid pays hourly at one-eighth of the computed 8-hour-style rate. ⁶⁸
- Compute funding off **mark-based** position value where the venue does so. Binance, Bybit, OKX, Coinbase, Deribit, and Hyperliquid all tie the relevant risk/funding state to mark rather than last. ⁶⁹
- Support interval changes and caps/floors. OKX and Bybit explicitly reserve the right to change funding frequency under stress; Hyperliquid has a hard cap; Binance and Deribit have clamp/cap logic in the official formulas. ⁷⁰

Liquidation simulation

- Use **mark price**, not last price, as the liquidation trigger unless the venue explicitly says otherwise. That is the dominant rule across the exchanges covered here. ⁷¹
- Include close-fee / taker-fee terms inside liquidation-price calculations when the venue does. Bybit's published formulas do this explicitly; OKX and Deribit charge forced-liquidation-related fees; Binance surfaces a symbol-level liquidation-fee field. ⁷²
- Support **staged liquidation** rather than immediate bankruptcy when the venue does partial reductions or pre-liquidation order cancellation. This is explicit on Coinbase INTX and described in various forms on OKX, Bybit UTA, and Hyperliquid. ⁷³
- Add insurance-fund / ADL / backstop branches. Binance, Bybit, OKX, Coinbase, Deribit, and Hyperliquid each document some nontrivial post-liquidation path when losses exceed immediate liquidation proceeds. ⁷⁴

Fee model

- Track maker and taker separately, by tier, by product family, and by whether the fill is opening, closing, liquidation-related, settlement-related, or delisting-related.
- Do not assume maker is always nonnegative or always better in realized strategy terms: some venues have rebates in some regimes; some allow special fee programs; some delisting settlements override ordinary fee tiers. ⁷⁵
- Model stop-market orders as taker-like executions and stop-limit orders as conditional entries that may miss or partially fill. The exchange docs show trigger price and execution price are not the same object. ⁷⁶

Contract lifecycle and symbol mapping

- Store **launch time / onboard date / creation timestamp** and **delivery or delisting time** per instrument. Binance, Bybit, OKX, and Deribit all publish these fields. Coinbase FIX instruments also support point-in-time add/modify/delete. ⁷⁷
- Preserve status transitions such as `PreLaunch`, `preopen`, `live`, `pause`, `reduce-only`, and `delete`. A backtest that treats a symbol as continuously tradable from the first bar to the last will create non-existent fills. ⁷⁸
- Handle venue-specific settlement and delisting procedures. Bybit can apply a 0.05% delisting settlement fee; OKX can auto-settle without fees and set funding to zero at the final settlement window; Hyperliquid builder perps can settle to current mark through `haltTrading`. ⁷⁹

Order-trigger semantics

- Store a separate field for **trigger reference price** where applicable: mark, last/contract price, or index. Binance, Bybit, OKX, Deribit, and Hyperliquid all document exchange-specific trigger sources. ⁸⁰
- Simulate venue protections such as Binance `triggerProtect`, Hyperliquid TP/SL slippage tolerance, and the distinction between live TPSL logic and regular stop logic on Coinbase FIX. ⁸¹

Unsafe backtest assumptions for small-account strategies

Ignoring funding is unsafe because it applies to notional, not posted margin. Using Binance's default interest component alone as a simple lower-bound illustration, **0.01% per 8h** becomes roughly **0.03% per day** on notional. On a \$10,000 position backed by \$1,000 margin, that is about **\$3/day**, or **0.3% of equity per day**, even before the premium component. Coinbase's hourly INTX funding and Hyperliquid's hourly settlement make this even more path-dependent intraday. Strategies with holding periods of hours to days can flip sign if funding is omitted. ⁸²

Using last price for liquidation is unsafe because the exchanges generally do not liquidate on last price. Binance explicitly warns that a position displayed with a liquidation threshold of **17,006 USDT** may be liquidated at a mark of **17,100** in a fast move. Bybit explicitly warns that a stop triggered on LTP can miss while liquidation is triggered on mark. A backtest that liquidates on last price can therefore either overstate survivability or trigger false liquidations, depending on the venue and the market state. ⁸³

Assuming flat commissions is unsafe because maker/taker spreads are not small relative to many short-horizon edges. A two-taker round trip at **5 bps per side** is **10 bps** before slippage. On Bybit's retail example rates, a taker entry and maker exit is **7.5 bps** round trip; two takers are **11 bps**. On Binance and OKX example rates, taker-taker is **10 bps** and maker-taker is **7 bps**. On Deribit, taker-taker is **10 bps**, while maker-taker can be materially lower because maker is **0.00%** in the retrieved examples. If your expected edge per trade is sub-10 bps, fee realism is not a refinement; it is the strategy. ⁸⁴

Ignoring liquidation fees and close-fee terms is unsafe because some venues explicitly bake them into liquidation math or charge them separately. Bybit's published liquidation formulas include **Estimated Fee to Close Position**. OKX charges forced-liquidation fees at the user's current taker tier. Deribit charges an **additional liquidation fee** to the insurance fund. Binance exposes **symbol-level liquidation fees**. A backtest that assumes liquidation simply exits at mark with the same ordinary fee as a normal close will systematically overstate residual equity after severe losses. ⁷²

Ignoring partial liquidations is unsafe because small accounts often sit near constraint boundaries where staged risk controls matter most. Coinbase INTX is explicit: below maintenance margin it partially liquidates back toward initial margin; only below close-out margin does the full waterfall begin. Hyperliquid routes most liquidations to the book and then escalates to backstop / ADL only if necessary. OKX and Bybit also describe staged or conditional halting behavior. Modeling everything as an immediate full-close exaggerates ruin frequency and distorts post-shock capital trajectories. ⁸⁵

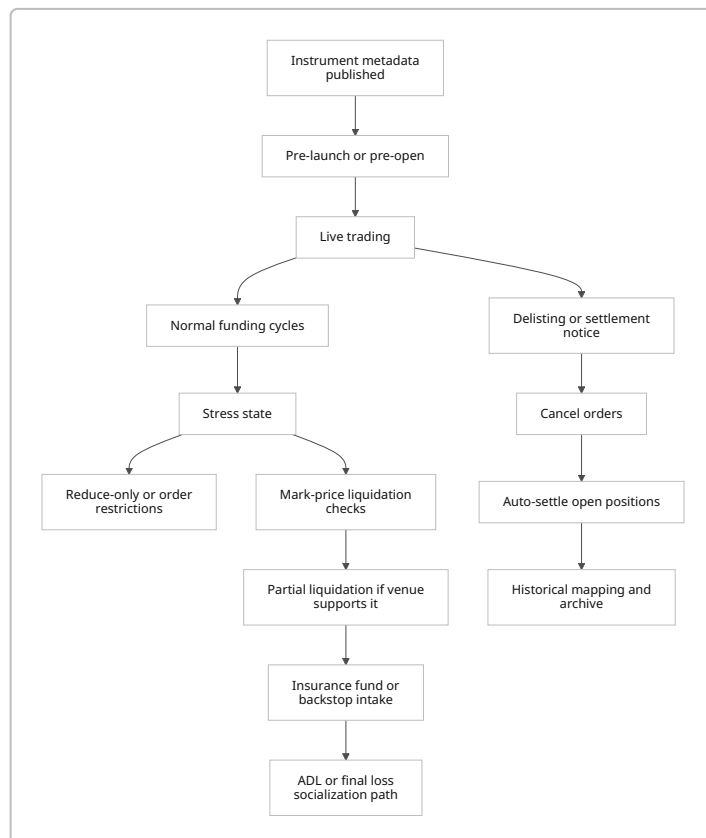
Assuming no delistings or lifecycle events is unsafe because venue behavior at delisting is not neutral. Bybit charges a **0.05%** fixed delisting settlement fee. OKX says delisted positions are auto-settled **without fee**, and the current funding rate at the relevant final window is set to **0**. Hyperliquid builder-deployed assets can be halted and settled to current mark. If a backtest ignores these cases, it will overstate capacity in tails and mis-handle PnL around the end of a contract's effective life. ⁷⁹

Assuming infinite liquidity is unsafe because multiple exchange formulas depend on **impact bid/ask** rather than top-of-book prints. Binance, Bybit, and OKX all use impact-price-style inputs in funding or mark/fair-value logic. A strategy that fills its own orders off top-of-book while also borrowing fund/mark logic from impact-based formulas is internally inconsistent. This matters most for small accounts in thin alts or for

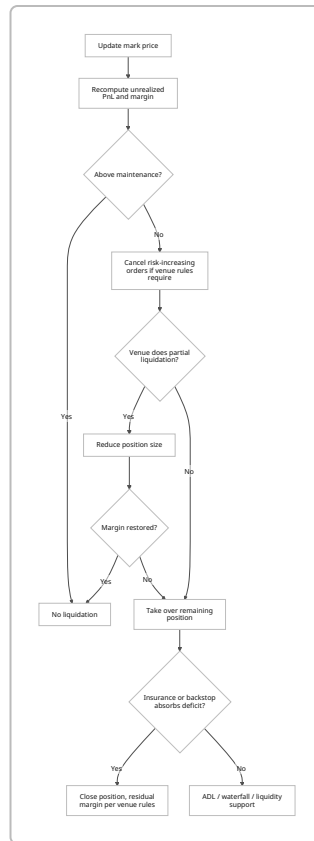
trigger orders placed during stress, where the difference between top-of-book, impact price, mark price, and actual stop execution can be large. ⁸⁶

Lifecycle and liquidation flow models

The lifecycle model below is the backtest state machine that is closest to what the retrieved exchange docs imply for modern crypto perps. It differs from a stock model because listing, pre-launch, live trading, reduce-only, delisting, and automatic settlement are first-class states rather than edge cases. ⁸⁷



The liquidation flow below captures the minimum logic a simulator should implement before it is trusted with leverage. The details differ across venues, but the broad structure—mark-based solvency test, staged engine action, then insurance/backstop/ADL escalation—is common. ⁸⁸



Open questions and limitations

This report is strongest where the official docs were explicit and weakest where official pages existed but I did not extract every detail before finalization. The main gaps are these. The gaps do **not** invalidate the core conclusions above, but they do matter for production-grade exchange simulators. ⁸⁹

For **Coinbase International Exchange**, the retrieved official documentation clearly covered funding, mark price, liquidation waterfall, market-data OI, and instrument state updates, but I did not extract the exact current **maker/taker tier table** from the official fees page, nor a separate public **REST OI history** endpoint. The FIX and help docs are still enough to conclude that INTX cannot be modeled like equities and that mark/funding/lifecycle are exchange-defined. ⁹⁰

For **Hyperliquid**, I retrieved the official funding, mark, liquidation, ADL, TP/SL, and settlement mechanics, but I did not extract the full **base fee tier schedule** numerically from the official fee page, and I did not pull the full contents of the main-venue **Delisting** page. The official docs still make two critical backtest points unambiguous: mark-driven risk logic is central, and lifecycle settlement can happen through protocol-level actions such as `haltTrading`. ⁹¹

For **Deribit**, the official support and API material retrieved here were sufficient on funding examples, fees, liquidation-fee existence, OI units, and lifecycle metadata, but I did not fully extract the latest support article's **complete mark-price sub-formula** line by line. The retrieved support material still confirms the

operational point that Deribit's mark is a fair-value-style reference distinct from last trade and that it governs PnL, margin, and liquidation. 92

1 3 7 66 67 What Are Mark Price and Price Index in USDⓈ-Margined Futures? | Binance Futures,Binance USDM,Binance Mark Price,Price Index USDM,Mark Price USDM

<https://www.binance.com/en/support/faq/detail/360033525071>

2 5 6 8 68 69 82 86 Introduction to Binance Futures Funding Rates | Binance Futures,What is funding rate,Binance Futures Funding Rates

<https://www.binance.com/en/support/faq/detail/360033525031>

4 24 26 71 83 88 Binance Futures Liquidation Protocols | Futures last price,Futures mark price,Binance futures liquidation,Mark price vs last price

<https://www.binance.com/en/support/faq/detail/360033525271>

9 11 Introduction to Funding Rate - Help Center

<https://www.bybit.com/en/help-center/article/Introduction-to-Funding-Rate>

10 Mark Price (Perpetual and Expiry Contracts) - Help Center

<https://www.bybit.com/en/help-center/article/Mark-Price-Calculation-Perpetual-Expiry-Contracts>

12 14 70 Perpetual funding fee mechanism | OKX

<https://www.okx.com/help/perps-funding-fee-mechanism>

13 Mark price and Last price | OKX

<https://www.okx.com/help/ii-mark-price-and-last-price>

15 17 90 How is the funding rate determined? | Coinbase Help

<https://help.coinbase.com/en/international-exchange/funding/how-is-the-funding-rate-determined>

16 54 55 57 INTX FIX Market Data Messages

https://docs.cdp.coinbase.com/international-exchange/fix-api/market-data?utm_source=chatgpt.com

18 20 92 Funding Specifications

https://support.deribit.com/hc/en-us/articles/31424939178397-Funding-Specifications?utm_source=chatgpt.com

19 Mark Prices

https://support.deribit.com/hc/en-us/articles/25944746962973-Mark-Prices?utm_source=chatgpt.com

21 23 91 Funding - Hyperliquid Docs - GitBook

https://hyperliquid.gitbook.io/hyperliquid-docs/trading/funding?utm_source=chatgpt.com

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https://hyperliquid.gitbook.io/hyperliquid-docs/trading/robust-price-indices?utm_source=chatgpt.com

25 Binance Futures Fee Structure & Fee Calculations | Binance Futures,Binance Fee Structure,Binance Fee Calculations

<https://www.binance.com/en/support/faq/detail/360033544231>

27 29 72 Liquidation Price Calculation under Isolated Mode (Unified Trading Account) - Help Center

<https://www.bybit.com/en/help-center/article/Liquidation-Price-Calculation-under-Isolated-Mode-Unified-Trading-Account>

28 79 84 Futures Contracts: Fees Explained - Help Center

<https://www.bybit.com/en/help-center/article/Perpetual-Futures-Contract-Fees-Explained>

30 32 **System Liquidation Mechanism | OKX**

<https://www.okx.com/help/vi-system-liquidation-mechanism>

31 75 **Trading Fee Rules FAQ | OKX**

<https://www.okx.com/help/trading-fee-rules-faq>

33 35 **What are the margin requirements and how do they apply to liquidations? | Coinbase Help**

<https://help.coinbase.com/en/international-exchange/margin-requirements/what-are-margin-requirements-and-how-do-they-apply-to-liquidations>

34 89 **International Exchange Fees | Coinbase Help**

<https://help.coinbase.com/en/international-exchange/trading-deposits-withdrawals/international-exchange-fees>

36 38 **Liquidations**

https://support.deribit.com/hc/en-us/articles/25944769313309-Liquidations?utm_source=chatgpt.com

37 **Fees**

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